DATA STRUCTURE AND PROGRAM DESIGN LAB – 07

7. Consider the undirected graph G, consisting of n nodes laid out in a 3-by- 3 grid: Start searching at node 1, and break ties for exploring the next node based on lower numerical order (i.e. add nodes to a queue low to high, add nodes to a stack high to low). (a) In what order are nodes marked as explored by BFS? (b) In what order are nodes marked as explored by DFS?

```
1. Search
2. Insert
3. Delete Leaf
4. Exit
Enter choice: 2
Enter element to insert: 10


1. Search
2. Insert
3. Delete Leaf
4. Exit
Enter choice: 2
Enter element to insert: 5
```

```
1. Search
2. Insert
3. Delete Leaf
4. Exit
Enter choice: 1
Enter element to search: 5
Found


1. Search
2. Insert
3. Delete Leaf
4. Exit
Enter choice: 4
```